

CROP CULTIVATION GUIDE FOR FARMERS

Practical Guidance for Planning, Establishing, Managing and Harvesting Major Crops

Introduction

Crop cultivation requires decisions at every stage, beginning with the selection of a suitable crop and continuing through land preparation, seed selection, sowing or planting, irrigation, nutrient management, weed control, pest and disease management, harvesting and post-harvest handling. The performance of a crop depends not only on the crop itself but also on soil type, climate, water availability, season, planting material, local pest and disease pressure and the management practices followed by the farmer.

This guide provides general cultivation information for commonly grown agricultural and horticultural crops. The recommendations are intended to support farm-level decision making and should be adapted to local soil conditions, season, irrigation facilities and officially recommended agricultural practices. Farmers should verify crop-specific fertilizer quantities, pesticide recommendations, seed rates and other technical recommendations with the relevant agricultural department, agricultural university, extension officer or locally applicable official guidance before making input decisions.

1. Crop Selection and Farm Planning

Crop selection should be based on the combination of soil characteristics, climate, water availability, season, available labour, expected production period, local market demand and the farmer's available resources. A crop that performs well in one location may not perform similarly in another location because soil, rainfall, temperature and pest pressure can differ considerably. Farmers should therefore consider both agronomic suitability and farm-level feasibility before selecting a crop.

Crop planning should also consider crop rotation and diversification. Growing the same crop continuously in the same field can increase the risk of certain pests, diseases, nutrient imbalance and soil-related problems. Alternating crops with different nutrient requirements and root systems can support soil health and reduce some recurring production problems. Farmers may also consider combining food crops, pulses, oilseeds, vegetables, fruits or other suitable crops according to their land and market conditions.

2. Paddy (Rice) Cultivation

Paddy, or rice, is an important cereal crop cultivated under irrigated, rainfed and different water-management systems. Successful paddy cultivation depends on selecting a variety suited to the local season, soil and water conditions, establishing a healthy crop through an appropriate nursery, transplanting or direct-seeding method, and managing water, nutrients, weeds, pests and diseases throughout crop development.

For transplanted paddy, a healthy nursery should be established using quality seed and suitable nursery management practices. The main field should be prepared so that the soil has adequate moisture and a suitable structure for transplanting. Seedlings should be transplanted at an appropriate age and spacing according to the selected variety and locally recommended cultivation method. Direct-seeded rice requires careful land preparation, uniform seed placement

and early weed management because weeds can compete strongly with young rice plants.

Water management is an important component of paddy cultivation. The field should not automatically be maintained under deep standing water throughout the entire crop cycle. Water management should be adjusted according to soil condition, crop growth stage, rainfall and the cultivation method being used. Excessive or poorly managed water can increase production costs and may create conditions favourable for certain problems, while insufficient water can reduce crop establishment and yield.

Paddy requires balanced nutrient management rather than excessive application of a single fertilizer. Nutrient recommendations should be based on soil condition, crop requirement and locally recommended fertilizer schedules. Nitrogen, phosphorus, potassium and other nutrients should be supplied according to crop demand and appropriate timing. Weed, insect and disease management should be based on regular field observation and integrated management practices.

Paddy harvesting should be carried out when the crop reaches the appropriate maturity stage and grain moisture is suitable for harvesting and subsequent handling. Harvesting too early or too late can increase grain losses and affect quality. After harvest, paddy should be handled carefully, dried to a suitable moisture level and stored under conditions that protect it from moisture, insects, rodents and contamination.

3. Wheat Cultivation

Wheat is an important cereal crop generally grown during a cool growing season and is used for food, feed and other products. Successful wheat production depends on selecting a variety suited to the local agro-climatic conditions, using quality seed, preparing a suitable seedbed and providing appropriate irrigation and nutrient management.

The field should be prepared to provide a reasonably fine and level seedbed with sufficient moisture and good drainage. Wheat should be sown during the locally recommended planting window because temperature and crop duration strongly influence establishment, tillering and grain development. Seed should be placed at an appropriate depth and spacing according to the selected variety and local cultivation recommendation. Uniform establishment helps the crop compete with weeds and supports more consistent grain development.

Wheat requires balanced nutrient management. Soil testing should be used where available to guide fertilizer decisions, while nitrogen, phosphorus, potassium and other nutrients should be managed according to crop requirement and local recommendations. Nitrogen should be supplied at appropriate stages rather than being applied excessively at one time. Excessive nitrogen can increase vegetative growth and may contribute to lodging or other management problems under some conditions.

Irrigation should be scheduled according to soil moisture, rainfall and crop growth stage. Water availability can be particularly important around stages associated with tillering, flowering and grain development, although the exact schedule depends on the production environment and soil. Excessive irrigation should be avoided because it can waste water, increase nutrient losses and create unfavourable root-zone conditions.

Wheat fields should be monitored for weeds, insect pests and diseases. Farmers should inspect leaves and stems for abnormal spots, discoloration, rust-like symptoms, insect feeding or other changes. Integrated management should combine preventive practices, suitable varieties where

available, field sanitation, monitoring and locally recommended interventions. Harvesting should be carried out when the crop reaches suitable maturity and grain moisture is appropriate for harvesting and storage. Proper drying and clean storage help protect grain quality and reduce losses.

4. Maize Cultivation

Maize is cultivated for food, feed and industrial purposes and can perform well when soil fertility, moisture and plant population are properly managed. Before planting, farmers should select a suitable hybrid or variety based on the local growing season, climate, intended use and availability of quality seed.

Land preparation should provide a reasonably uniform seedbed with sufficient soil moisture and good drainage. Maize is sensitive to severe waterlogging, particularly during early establishment and later reproductive stages. Therefore, fields with poor drainage should be corrected before planting. Seed should be placed at an appropriate depth and spacing according to the recommended production system.

Nutrient management is important because maize can respond strongly to adequate nutrient availability. Fertilizer application should be based on soil-test information and local recommendations. Nitrogen should be managed carefully because both deficiency and excessive application can affect crop performance. Adequate phosphorus and potassium are also important for root development, plant strength and grain formation.

Weed competition can significantly affect young maize plants. The field should therefore be monitored during the early growth period and weeds should be managed using appropriate cultural, mechanical or other locally approved methods. Farmers should regularly inspect maize for insect damage, leaf symptoms, stalk problems and ear-related issues. Integrated pest and disease management should be followed instead of depending only on chemical control.

Maize should be harvested when the crop reaches physiological maturity and the grain has developed adequately. Harvest timing should consider grain moisture, intended use and local drying facilities. Proper drying and storage are important to prevent fungal growth, insect infestation and quality deterioration.

5. Groundnut Cultivation

Groundnut is an important oilseed and food crop that requires suitable soil conditions for proper peg penetration and pod development. Well-drained sandy loam or other suitable soils with appropriate physical properties are generally preferred. Waterlogging can adversely affect root development and pod formation, so drainage should be considered before planting.

Quality seed should be selected from a reliable source and should be suitable for the local season and production environment. Seed treatment, where recommended by agricultural authorities, can support healthy crop establishment. Sowing should be carried out at an appropriate time with suitable spacing and soil moisture so that plants establish uniformly.

Groundnut requires careful nutrient and moisture management. Calcium and other nutrients can be important for pod development, while overall nutrient management should be based on soil testing and local recommendations. Irrigation should be scheduled according to crop requirement, rainfall and soil condition. Excessive irrigation can create waterlogging and increase the risk of

certain diseases.

Weed control is particularly important during early crop growth because weeds compete for water, nutrients and sunlight. Farmers should monitor the field regularly and use appropriate integrated weed management practices. Groundnut should also be inspected for leaf spot diseases, rust, collar rot, root-related problems and insect damage. Disease management should focus on clean seed, crop rotation, field sanitation, appropriate irrigation and other integrated measures.

Harvesting should be done when pods have reached suitable maturity. Delayed harvesting can increase the risk of pod loss and deterioration, while premature harvesting can reduce yield and quality. After lifting the plants, pods should be dried properly before storage to reduce the risk of fungal growth and quality loss.

6. Cotton Cultivation

Cotton requires a suitable growing period, adequate sunlight, appropriate soil conditions and careful management of water and nutrients. Farmers should select varieties or hybrids recommended for the local production environment and should obtain quality seed from reliable sources.

The field should be prepared to provide good root development and drainage. Cotton does not perform well when roots remain in saturated soil for prolonged periods. Sowing should be carried out according to the recommended local season, spacing and planting method. Proper plant population is important because overcrowding can increase competition and may influence pest and disease conditions.

Nutrient management should be based on soil testing and crop requirement. Nitrogen, phosphorus, potassium and micronutrients may all contribute to crop development, but excessive fertilizer application should be avoided. Irrigation should be managed according to soil moisture, rainfall and crop growth stage, with particular attention to periods when water demand increases.

Cotton fields require frequent monitoring for sucking pests, boll-related pests and disease symptoms. Farmers should inspect leaves, shoots, squares and bolls rather than waiting until visible damage becomes severe. Integrated pest management can include field sanitation, resistant or tolerant varieties where available, biological methods, monitoring and locally approved interventions when necessary.

Cotton harvesting should generally be carried out when bolls open properly and lint is sufficiently dry. Harvested cotton should be kept clean and protected from moisture, soil and other contaminants because contamination can reduce fibre quality and market value.

7. Tomato Cultivation

Tomato is a widely cultivated vegetable crop that can be grown under suitable climatic conditions in open fields or protected systems. Successful production depends on selecting a suitable variety, producing healthy seedlings, preparing the field properly and maintaining balanced irrigation and nutrient management.

Healthy seedlings should be established in a suitable nursery environment and transplanted when they have developed adequately. The main field should have good drainage because prolonged waterlogging can damage roots and increase disease risk. Plant spacing should be selected according to variety, growth habit and cultivation system.

Tomato requires regular moisture, but irrigation should be managed carefully. Irregular water supply can affect flowering, fruit development and fruit quality, while excessive moisture can contribute to root and disease problems. Nutrient application should be based on soil condition and crop requirement, with balanced nutrition throughout the growth period.

Tomato plants should be inspected regularly for leaf spots, wilting, viral-like symptoms, fruit damage and insect infestation. Common problems can include fungal, bacterial and viral diseases as well as insect pests such as fruit borers, whiteflies, thrips and other sap-feeding insects. Since similar symptoms can have different causes, farmers should avoid assuming a diagnosis based only on one visible symptom.

Harvesting should be carried out according to the intended market and transport distance. Fruits may be harvested at different maturity stages depending on whether they are intended for immediate local sale or transportation. Careful handling is important because bruising can accelerate deterioration.

8. Chilli Cultivation

Chilli is cultivated for both fresh consumption and dry chilli production. Crop performance depends on selecting a suitable variety, maintaining healthy seedlings and providing appropriate soil moisture and nutrient management. Fields should have adequate drainage because prolonged waterlogging can damage roots and increase disease pressure.

Seedlings should be raised using quality seed and transplanted at an appropriate stage. Irrigation should be regular but should not create prolonged water saturation. Balanced nutrient management is important for vegetative growth, flowering and fruit production. Excessive nitrogen can promote excessive vegetative growth and may not always result in better fruit production.

Chilli should be monitored for thrips, mites, aphids, fruit-related pests and disease symptoms. Leaf curling, discoloration, stunting and flower or fruit damage can arise from different causes, including insect feeding, viral infections, nutritional problems or environmental stress. Therefore, diagnosis should consider multiple symptoms and field conditions rather than relying on a single visual sign.

For dry chilli production, harvesting and drying require special attention. Fruits should be harvested at the appropriate maturity stage and dried under hygienic conditions that protect them from moisture, contamination and fungal growth.

9. Brinjal Cultivation

Brinjal is a vegetable crop that can be grown across suitable warm environments. Healthy seedlings and good field preparation are important for successful establishment. The field should be well drained and should provide sufficient space for root and canopy development.

Irrigation should maintain adequate soil moisture without prolonged waterlogging. Nutrient management should be balanced and based on soil fertility and crop requirements. Organic matter and appropriate nutrient management can support soil structure and crop development.

Brinjal is susceptible to several insect pests and diseases. Farmers should regularly inspect the growing tips, leaves, stems and fruits for feeding damage, wilting, discoloration and abnormal growth. Shoot and fruit borer damage can reduce marketable yield, while wilt and other diseases

may cause plant decline. Integrated management should include healthy planting material, field sanitation, removal of severely affected plant parts where appropriate, monitoring and locally recommended control measures.

Fruits should be harvested at the appropriate tender stage according to market requirements. Frequent harvesting can encourage continued production in suitable varieties and conditions. Harvested fruits should be handled gently to reduce mechanical damage.

10. Onion Cultivation

Onion production can be carried out through seedlings or other locally recommended establishment methods depending on the production system. The crop requires suitable soil, good drainage and careful irrigation because excessive moisture can contribute to bulb and root problems.

Healthy planting material is important for uniform crop establishment. Nutrient management should consider soil fertility and the crop's requirements during vegetative growth and bulb development. Excessive nitrogen late in the crop cycle can affect bulb maturity and storage characteristics, so fertilizer timing should follow local recommendations.

Onion fields should be monitored for thrips, fungal diseases, bacterial problems and other causes of leaf damage. Irrigation should be reduced or managed appropriately as bulbs approach maturity according to local production practices and weather conditions. Harvesting should be carried out when bulbs reach suitable maturity and tops show appropriate signs of maturity.

Proper curing is important after harvest. Bulbs should be dried under suitable conditions so that the outer layers become sufficiently dry before storage. Storage areas should be clean, dry and well ventilated.

11. Potato Cultivation

Potato requires suitable soil conditions that allow tuber development and harvesting. Loose, well-drained soil is generally important because compacted or waterlogged conditions can restrict tuber growth and increase disease risk. Quality, healthy planting material should be selected from a reliable source.

Planting depth and spacing should be appropriate for the selected variety and production system. Earthing-up or other locally recommended practices may be used to support tuber development and protect developing tubers from unsuitable exposure. Irrigation should be carefully scheduled because both drought stress and excessive moisture can affect crop development.

Potato fields should be monitored for foliage diseases, tuber problems and insect pests. Farmers should pay attention to unusual leaf spots, wilting, discoloration and insect feeding. Harvesting should be performed when tubers have reached suitable maturity and skin development is adequate for the intended market or storage system.

12. Sugarcane Cultivation

Sugarcane is a long-duration crop requiring substantial planning for land preparation, planting material, irrigation, nutrient management and harvesting. Farmers should select varieties suited to the local climate, soil and production conditions and should obtain healthy planting material from reliable sources.

The field should be prepared adequately to allow root and cane development. Planting should be carried out during the locally recommended season using an appropriate planting method and spacing. Water management is particularly important because sugarcane requires adequate moisture during active growth but can be affected by prolonged waterlogging.

Nutrient management should be based on soil testing and crop requirements. Nutrients should be supplied at appropriate stages rather than relying on excessive application at one time. Weed management is important during early crop establishment because weeds can compete strongly with young sugarcane.

Sugarcane should be monitored for stem and leaf pests, fungal diseases and other crop problems. Crop residues and field sanitation should be managed appropriately, and crop rotation or other preventive practices may be considered where suitable. Harvesting should be carried out at the appropriate maturity stage according to the variety, local practice and intended processing requirements.

13. Banana Cultivation

Banana is a perennial or long-duration fruit crop that requires suitable temperature, soil, drainage and water availability. Healthy planting material, such as quality suckers or suitable tissue-culture plants from reliable sources, is important for establishing a uniform plantation.

The field should have good drainage because banana roots can be damaged by prolonged waterlogging. Regular irrigation is often required depending on rainfall and soil conditions, but irrigation should be adjusted to avoid excessive moisture. Nutrient requirements can be significant, so fertilizer management should be based on soil condition, plant age and locally recommended practices.

Banana plantations should be monitored for leaf diseases, rhizome and root problems, insect pests and physiological disorders. Removing severely affected plant material and maintaining field sanitation can help reduce disease pressure. Plants should be supported where necessary in locations prone to wind damage, especially when bunches become heavy.

Bunches should be harvested according to the maturity required for the target market. Harvesting and transportation should minimize bruising because banana fruit can be easily damaged during handling. Clean storage and appropriate temperature management can help maintain quality.

14. Coconut Cultivation

Coconut is a long-lived plantation crop that requires long-term management of soil fertility, water, pests and diseases. Planting material should be selected from reliable sources, and planting sites should have suitable soil depth and drainage.

Young coconut palms require appropriate irrigation and nutrient management during establishment. Mature palms also require continued attention to soil fertility and moisture, particularly during periods of low rainfall. Organic matter management and suitable intercrops may be considered where locally appropriate.

Coconut palms should be inspected regularly for leaf, crown, trunk and fruit-related symptoms. Pest and disease management should focus on early detection, field sanitation, removal of affected material where recommended and other integrated approaches. Farmers should obtain expert guidance when serious crown or trunk symptoms are observed.

Harvesting should be carried out according to the intended use, such as tender coconut or mature nuts. Climbing and harvesting should be conducted using safe equipment and trained workers because coconut harvesting involves significant fall hazards.

15. Pulses and Legume Crops

Pulse crops such as green gram, black gram, red gram and other locally suitable legumes can contribute to food production and may also support soil fertility through biological nitrogen fixation. Crop selection should consider the season, soil type, rainfall and duration of the variety.

Good-quality seed and proper crop establishment are important for achieving a uniform plant population. Excessive irrigation should be avoided where the crop is sensitive to waterlogging. Weed management is important during early crop growth, while nutrient management should consider soil fertility and the specific nutrient requirements of the selected pulse crop.

Pulse crops should be monitored for pod borers, sucking pests, fungal diseases, viral-like symptoms and other field problems. Farmers should inspect leaves, flowers and pods regularly and use integrated management practices. Harvesting should be timed to reduce shattering and losses, particularly when pods mature unevenly.

16. Crop Rotation and Sequential Cropping

Crop rotation involves growing different crops in succession on the same land rather than repeatedly growing the same crop. Properly planned rotation can help manage some pests and diseases, improve nutrient use and support soil structure. Including legumes in suitable rotations may also contribute to biological nitrogen fixation.

Sequential cropping allows farmers to use the same field for more than one crop during different periods of the year. The sequence should be selected according to the local climate, irrigation availability, crop duration and market conditions. Crop residues, nutrient requirements and disease carryover should be considered when designing the sequence.

17. Irrigation Management Across Crops

Irrigation should be based on crop requirement, soil moisture, rainfall, weather conditions and growth stage. Different crops have different water requirements, and even the same crop may require different amounts of water at different stages. Farmers should therefore avoid using a fixed irrigation schedule without considering actual field conditions.

Over-irrigation can waste water, increase production costs, cause nutrient losses and create favourable conditions for certain root and soil-borne diseases. Under-irrigation can cause water stress and reduce growth, flowering, fruit development or grain filling. Efficient irrigation methods and proper scheduling can help improve water-use efficiency.

18. Fertilizer and Nutrient Management

Crop nutrition should be based on soil testing wherever possible. Fertilizer decisions should consider the crop, variety, soil fertility, expected yield, previous crop and local recommendations. Applying more fertilizer than required does not necessarily increase yield and may create economic and environmental problems.

Organic inputs such as compost, farmyard manure and other locally suitable materials can contribute to soil organic matter when they are properly prepared and applied. Chemical fertilizers and organic inputs can be combined as part of an integrated nutrient management approach when appropriate.

19. Weed Management

Weeds compete with crops for water, nutrients, sunlight and growing space. Weed competition is often particularly important during crop establishment and early growth. Farmers should identify the major weeds present in the field and select suitable management practices.

Weed management can include preventive measures, crop rotation, proper land preparation, manual or mechanical removal, mulching and other locally recommended methods. Chemical weed control, where appropriate, should use only registered products and label-compliant application practices. Farmers should not increase herbicide dosage or frequency without official guidance.

20. Pest and Disease Monitoring

Regular field monitoring is one of the most important components of crop management. Farmers should inspect the crop systematically rather than waiting until widespread damage becomes visible. Observations may include changes in leaf colour, spots, wilting, holes, insect presence, stem damage, flower loss, fruit damage and unusual plant growth.

A symptom does not always identify a specific disease or pest. For example, leaf yellowing may result from nutrient deficiency, water stress, root damage, disease or other environmental conditions. Correct diagnosis should therefore consider the crop, plant age, affected plant parts, pattern of symptoms, recent weather, irrigation, fertilizer application and presence of insects.

21. Harvesting and Post-Harvest Management

Harvesting should be carried out according to crop maturity and market requirements. Harvesting too early can reduce quality and yield, while delayed harvesting may increase field losses or deterioration. The appropriate harvesting method differs between grains, pulses, vegetables, fruits, oilseeds and plantation crops.

Post-harvest handling should protect agricultural produce from moisture, contamination, pests and physical damage. Produce should be cleaned, dried, sorted or graded where necessary. Storage conditions should be suitable for the crop, and stored produce should be inspected regularly for insects, fungal growth and other deterioration.

22. Crop-Specific Decision Making

Farmers should avoid applying the same cultivation method to every crop because crops differ in their soil, water, nutrient, temperature and pest requirements. A cultivation decision should consider the crop, variety, location, season, soil, irrigation availability and current field condition.

When a farmer asks an agricultural assistant about a crop, the answer should ideally connect the crop with its growth stage and the observed problem. For example, a recommendation for a newly transplanted vegetable crop may differ from the recommendation for the same crop during flowering or fruit development. This stage-based approach can improve the usefulness of farm decisions.

23. When Expert Advice Is Required

Farmers should seek professional agricultural advice when symptoms are severe, rapidly spreading, difficult to identify or associated with major crop loss. Expert support is also appropriate when farmers are uncertain about pesticide selection, fertilizer quantities, disease diagnosis, irrigation requirements or government-supported agricultural inputs.

Photographs and additional information can improve remote agricultural diagnosis, but an image alone may not always be sufficient. Information such as crop name, variety, crop age, location, soil type, irrigation method, recent weather, fertilizer use and the pattern of affected plants can help an agricultural expert assess the situation more accurately.

24. Using an Agricultural Assistant for Crop Management

An agricultural assistant can help farmers organize information about crop cultivation, identify possible causes of visible problems, understand general management practices and locate relevant agricultural guidance. The assistant can also connect a farmer's question with information from crop cultivation, disease management, general agriculture guidelines and government scheme documents.

The assistant should not present uncertain disease identification as a confirmed diagnosis. When a question lacks sufficient information, it should request relevant details such as crop, variety, age, symptoms, affected plant parts, field conditions and recent management practices.

Recommendations involving regulated agricultural chemicals, specific application rates or rapidly changing government schemes should be verified against current official guidance before implementation.

25. Overall Crop Cultivation Approach

Successful crop production is a continuous process rather than a single decision. Farmers should begin with appropriate crop and variety selection, prepare the field according to soil and water conditions, establish healthy plants, maintain balanced nutrition and irrigation, monitor weeds, pests and diseases, and harvest at the appropriate maturity stage.

Good crop management also requires regular observation and record keeping. Farmers can improve future decisions by recording sowing dates, varieties, irrigation events, fertilizer applications, pest observations, disease symptoms, harvesting dates and approximate production. Comparing these records across seasons can help identify recurring problems and improve farm planning.

Conclusion

Crop cultivation is most successful when management decisions are adapted to the specific crop, soil, season, weather and farm resources. General agricultural knowledge provides a useful foundation, but crop-specific and location-specific recommendations should be verified with reliable agricultural authorities and current technical guidance.

A farmer-focused agricultural assistant can support this process by combining information from cultivation guides, agriculture guidelines, disease and pest management resources and government scheme information. The purpose of such an assistant is to provide understandable, evidence-based guidance while recognizing uncertainty and directing farmers to appropriate expert or official sources when a decision requires specialized or current information.

Reference and Information Sources

Crop cultivation information should be developed and maintained using reliable agricultural sources such as the Indian Council of Agricultural Research, State Agricultural Universities, State Departments of Agriculture and Horticulture, Krishi Vigyan Kendras, agricultural research institutions and other official agricultural extension resources. Crop-specific recommendations should be reviewed periodically because recommended varieties, pest-management practices, input regulations and cultivation technologies can change over time.